

Day 3 — Naming the 15 Tithis

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will define *tithi* in terms of the Moon’s elongation from the Sun (12°), recite the 15 tithi names of a pakṣa in order, and explain why a tithi need not equal a 24-hour calendar day.

Materials

- Board with three columns ready: Number | IAST name | English (ordinal)
- A printed handout of the 15 tithi names (one per student)
- The white ball and torch from Day 1 (one quick re-use)
- Notebooks

Vocabulary introduced today

- *tithi* (IAST: tithi; lit. “lunar day”) — $1/30$ of a synodic month; precisely, the time the Moon takes to gain 12° of angular distance (“elongation”) from the Sun.
- *elongation* — the angle between the Sun and Moon as seen from Earth (0° at *amāvasyā*; 180° at *pūrṇimā*).

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What pakṣa is it?*” — 2 students share.
- Recall: write on the board — “29.5 days, 2 pakṣas, 2 anchor days.” Ask the class to fill in: how many days in a pakṣa? (~ 14.75 .) How many parts will we divide that into? (15.)

Core (20 min)

1. The tithi as a unit.

“A tithi is one-thirtieth of a full lunar cycle. So a synodic month has 30 tithis. Each pakṣa has 15.”

On the board:

Synodic month (29.53 days)

└─ Śukla pakṣa = 15 tithis (1 → 15)
└─ Kṛṣṇa pakṣa = 15 tithis (1 → 15)

2. The precise definition (this is important).

“A tithi is NOT one-thirtieth of a day. It is one-thirtieth of the cycle.”

The full cycle covers 360° of elongation (the Moon goes all the way around). Divide 360° by 30 — each tithi is 12° of elongation.

One tithi = the time during which the Moon gains 12° of elongation on the Sun.

On average, this takes about 23 hours and 37 minutes. But the Moon doesn't move uniformly — sometimes faster (perigee, closer to Earth) and sometimes slower (apogee, farther). So a single tithi can last anywhere from ~21.5 hours to ~26 hours.

3. Why a tithi is not a day — three consequences.

- Some tithis “skip” a sunrise. If a tithi starts at 7 a.m. and ends at 5 a.m. the next morning, it never overlaps with a sunrise. This is called *kṣaya* tithi (“decayed” — it never had a sunrise of its own).
- Some tithis “span” two sunrises. If a tithi starts at 5 a.m. on one day and ends at 7 a.m. two days later, it covers two sunrises. This is *vṛddhi* (“growth”).
- Most tithis are tagged to one calendar date — the date during which the tithi is current at sunrise.

4. The 15 names — they are Sanskrit ordinals.

On the board, build the list. Students copy. (Show *śukla* column; same names apply to *kṛṣṇa* with the prefix.)

#	IAST	English (ordinal)
1	<i>pratipadā</i>	“first”
2	<i>dvitīyā</i>	“second”
3	<i>ṛtīyā</i>	“third”
4	<i>caturthī</i>	“fourth”
5	<i>pañcamī</i>	“fifth”
6	<i>ṣaṣṭhī</i>	“sixth”
7	<i>saptamī</i>	“seventh”
8	<i>aṣṭamī</i>	“eighth”
9	<i>navamī</i>	“ninth”
10	<i>daśamī</i>	“tenth”
11	<i>ekādaśī</i>	“eleventh”
12	<i>dvādaśī</i>	“twelfth”
13	<i>trayaśī</i>	“thirteenth”
14	<i>caturdaśī</i>	“fourteenth”
15	<i>pūrṇimā</i> (or <i>amāvasyā</i>)	full / new moon

Note the pattern: 1–10 are simple ordinals; 11–14 are compound (*eka+daśī*, *dvā+daśī*, *trayaḥ+daśī*, *catur+daśī*); the 15th is always the anchor day — *pūrṇimā* at the end of *śukla*, *amāvasyā* at the end of *kṛṣṇa*.

5. Sanskrit number recall. *Eka*, *dvi*, *tri*, *catur*, *pañca*, *ṣaṣ*, *sapta*, *aṣṭa*, *nava*, *daśa* — count it together once. The tithi names will stick faster if students see them as ordinals first, Sanskrit terms second.

Activity (15 min) — Tithi drill

Two parts:

Part A (8 min) — paired recall. In pairs, partners alternate: one says a tithi number (1–15), the other names the tithi in IAST. Switch roles every minute. Goal: name all 15 in under 60 seconds by end of activity.

Part B (7 min) — date-to-tithi. Distribute a printed list of 5 dates and the tithi at sunrise on each (lookup done by the teacher from any standard *pañcāṅga* — e.g. for a sample week). Students answer: - *Is this date in śukla or kṛṣṇa pakṣa?* - *What's the Sanskrit name of the tithi?* - *What shape will the Moon be that night?*

Walk around. Check that students are reading the table, not guessing.

Wrap-up (5 min)

- **Formative quiz on a slip of paper (5 questions, 3 min)** — see quizzes/formative.md Part A.
- Collect for grading. Next day's class includes the result.
- **Preview Day 4:** “Tomorrow we build a tithi calendar for THIS calendar month.”

Student-facing content

A tithi is the time the Moon takes to gain 12° on the Sun.

- $360^\circ / 30 = 12^\circ$ per tithi.
- Average tithi length: ~23 h 37 min — close to a day but not exactly a day.
- Some tithis skip a sunrise (*kṣaya*); some span two (*vṛddhi*).

Each pakṣa has 15 tithis named as Sanskrit ordinals:

pratipadā, dvitīyā, tṛtīyā, caturthī, pañcamī, ṣaṣṭhī, saptamī, aṣṭamī, navamī, daśamī, ekādaśī, dvādaśī, trayodaśī, caturdaśī, pūrṇimā (or amāvasyā).

Homework

- Memorise the 15 tithi names. Quiz yourself or be quizzed at home.
- Continue the Moon diary — now naming the tithi as well as the pakṣa.

Differentiation

- **One tier up:** Look up the formula for tithi computation: $\text{tithi number} = \text{floor}((\text{Moon_ecliptic_longitude} - \text{Sun_ecliptic_longitude}) \bmod 360) / 12 + 1$. Use an astronomy app (Stellarium is free) to get the longitudes at any moment and compute the tithi yourself. Compare with a published *pañcāṅga*.
- **One tier down:** Memorise only the 6 anchor tithis — *pratipadā, pañcamī, aṣṭamī, ekādaśī, caturdaśī, pūrṇimā/amāvasyā*. Fill the rest in next week.

Teacher notes

- The 12°-per-tithi rule is the single most important computational fact in this module. Drill it.
- Watch for “tithi = day” confusion — every quiz will catch this. The simplest fix: ask “if a tithi started at 7 a.m. and the next started at 5 a.m. the next morning, how many hours was that tithi?” (22.) Repeat with different times until students see that tithi length varies.
- Pronunciation: *ṣaṣṭhī* is hard — “shash-tee.” *Ekādaśī* — “ay-KAH-da-shee.” Don’t rush.

Citation(s) used in this lesson

- Paraphrase from *Sūrya-siddhānta*, chapter 2: the tithi is defined as the period during which the Moon and Sun separate by 12° of arc. Cited by text name only; we do not quote a specific verse.